Time-Frequency and Wavelets:

- Until now, we focused on Fourier transforms (i.e. FS, CTFT, DTFT, DFT), which construct and deconstruct signals using sinusoids.
- But sinusoidal analysis has some serious limitations, as we will soon discuss.
- In this note packet, we will cover:
  - limitations of Fourier analysis
  - the short-time CTFT
  - the time/frequency uncertainty principle
  - the short-time DTFT
  - the short-time DFT
  - the continuous wavelet transform (CWT)
  - the discrete wavelet transform (DWT)
  - some applications of the DWT.

Limitations of Fourier Analysis:

- Consider the world of continuous-time signals $\{x(t)\}_{t\in\mathbb{R}}$
- For these signals, we often use the CTFT:

$$X_{\mathsf{c}}(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}\,\mathrm{d}t, \quad \Omega \in \mathbb{R}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_{\mathsf{c}}(j\Omega)e^{j\Omega t}\,\mathrm{d}\Omega, \quad t \in \mathbb{R}.$$

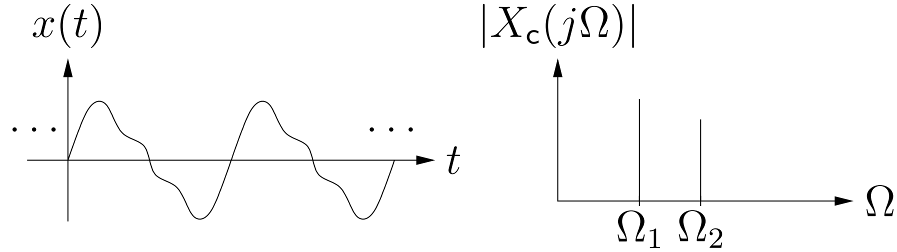
- The CTFT constructs $x(t)$ from basis elements $b_\Omega(t) \triangleq \frac{1}{\sqrt{2\pi}}e^{j\Omega t}$ and basis coefficients $\frac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)$:

$$x(t) = \int_{-\infty}^{\infty} \tfrac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)\, b_\Omega(t)\,\mathrm{d}\Omega$$

$$\tfrac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega) = \int_{-\infty}^{\infty} x(t)\, b_\Omega^*(t)\,\mathrm{d}t,$$

  where $\frac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)$ tells how much $b_\Omega(t)$ is present in $x(t)$.

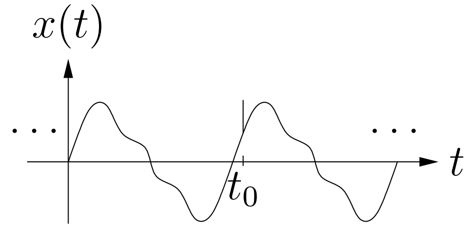
- When $\{x(t)\}_{t\in\mathbb{R}}$ is a sum of a few sinuoids, the CTFT $\{X_{\mathsf{c}}(j\Omega)\}_{\Omega\in\mathbb{R}}$ is very nice because it gives us a concise description of how the signal is constructed:

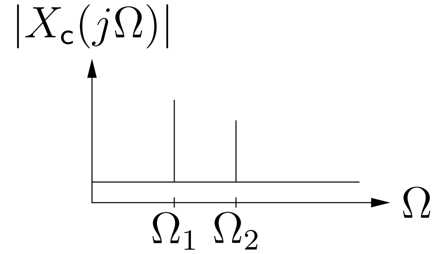


- But other times, the CTFT does not give the best "view."

Limitations of Fourier Analysis (cont.):

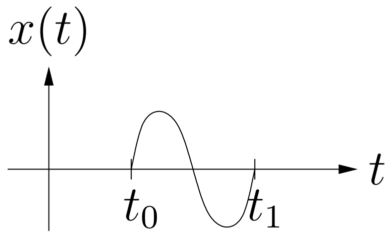
- For example, consider adding a sharp "glitch" (e.g., a Dirac delta) to the sum-of-sinusoids signal at time t_0. This adds $\{e^{-j\Omega t_0}\}_{\Omega\in\mathbb{R}}$ to the CTFT:

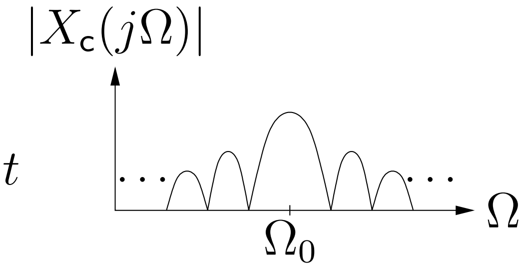




  The glitch info t_0 is better visible in the time domain.

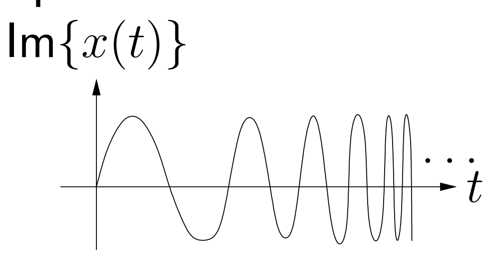
- Or consider a windowed version of $\{\sin(\Omega_0 t)\}_{t\in\mathbb{R}}$:

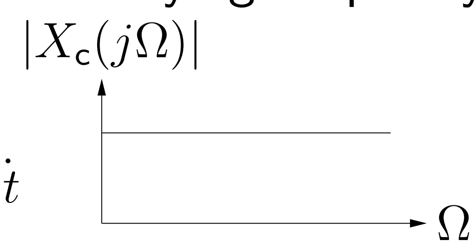




  Although $x(t)$ consists of a "pure" sinusoid over $t \in [t_0, t_1]$, this is not clearly visible from the CTFT.

- Or consider the complex linear chirp $x(t) = e^{j\Omega_0 t^2} = e^{j\Omega(t)t}$, a complex sinusoid with time-varying frequency $\Omega(t) = \Omega_0 t$



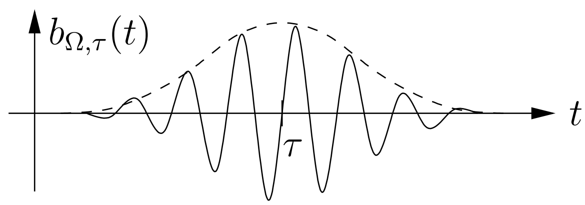


  The CTFT magnitude is flat; it doesn't show the interesting structure of the chirp signal.

The Short-Time CTFT:

- The problem with the CTFT is that "frequency" is *time-invariant*. That is, the CTFT constructs $x(t)$ from sinusoids $b_\Omega(t) = e^{j\Omega t}$ spread uniformly over all time t.
- What if we instead constructed a basis from *windowed sinusoids* localized at time $t \approx \tau$ (for many different τ):

$$b_{\Omega,\tau}(t) \triangleq \frac{1}{\sqrt{2\pi}} w(t-\tau) e^{j\Omega t} \text{ for } t \in \mathbb{R},$$



We'll assume real-valued $w(t)$ s.t. $\int_{-\infty}^{\infty} w(t)^2 \, \mathrm{d}t = 1$.

- We could use them to construct/deconstruct $x(t)$ via

$$x(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} X_{\mathsf{c}}(j\Omega, \tau) \, b_{\Omega,\tau}(t) \, \mathrm{d}\Omega \, \mathrm{d}\tau$$

$$\frac{1}{\sqrt{2\pi}} X_{\mathsf{c}}(j\Omega, \tau) = \int_{-\infty}^{\infty} x(t) b_{\Omega,\tau}^*(t) \, \mathrm{d}t \quad \leftarrow \text{"coefficients"}$$

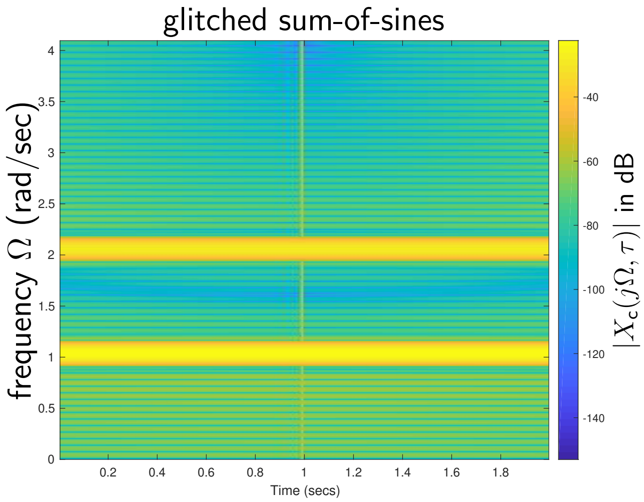
- After rearranging, we have the *short-time CTFT*:

$$X_{\mathsf{c}}(j\Omega, \tau) = \int_{-\infty}^{\infty} x(t) w(t-\tau) e^{-j\Omega t} \, \mathrm{d}t$$

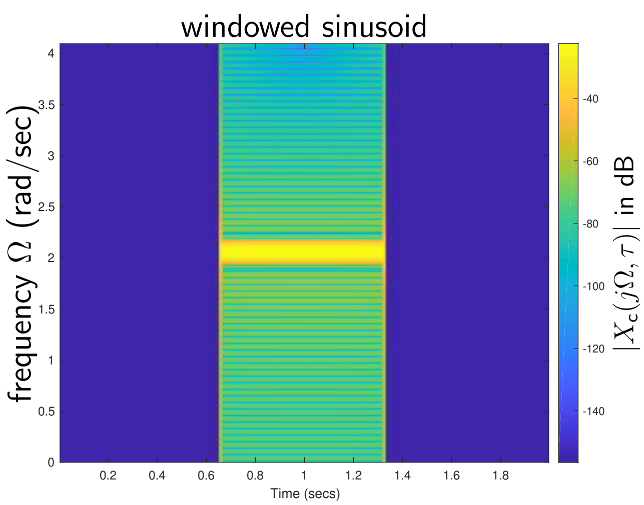
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X_{\mathsf{c}}(j\Omega, \tau) w(t-\tau) e^{j\Omega t} \, \mathrm{d}\Omega \, \mathrm{d}\tau.$$

ST-CTFT of Previous Examples:

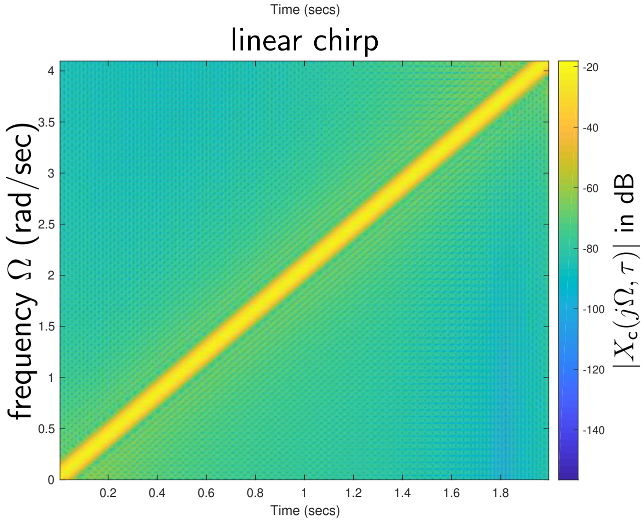
Using a window $w(t)$ of width $1/64$ sec...



Note sinusoids at $\Omega_1 = 1$ and $\Omega_2 = 2$, and glitch at time $\tau = 1$.



Note sinusoid at $\Omega_0 = 2$ windowed over time interval $\tau \in [\frac{2}{3}, \frac{4}{3}]$.

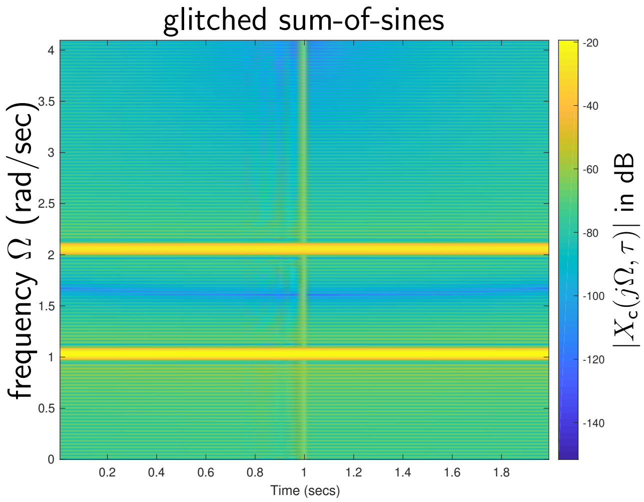


Note linear chirp with instantaneous frequency $\Omega(\tau) = \tau$.

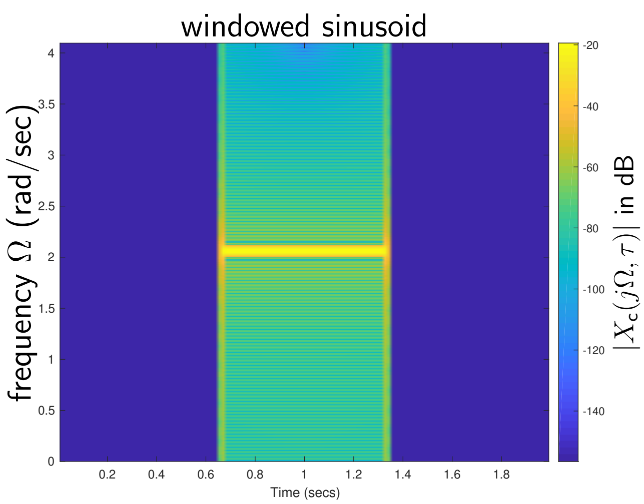
Note resolutions in time and frequency domains.

ST-CTFT of Previous Examples (cont.):

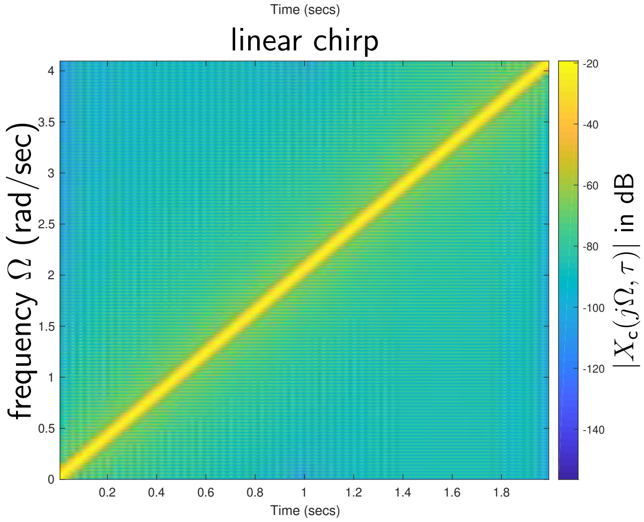
Using a wider window $w(t)$ of width $1/32$ sec...



Note sinusoids are sharper but glitch is more blurred.



Note sinusoid is sharper but edges are more blurred.



Similar to before.

Frequency resolution improved, but time resolution degraded!

Time and Frequency Resolution:

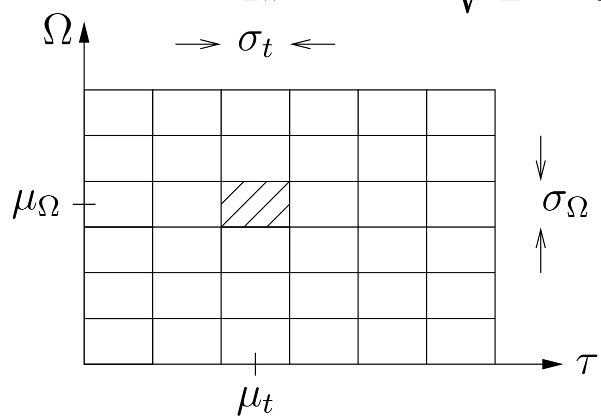
- The previous examples showed that, as the window $w(t)$ gets wider, the ST-CTFT frequency resolution improves but the time resolution degrades.
- Let's try to understand this more quantitatively.
- Consider an arbitrary basis element $b(t)$ with energy

$$E \triangleq \int_{-\infty}^{\infty} |b(t)|^2 \, \mathrm{d}t.$$

- Let's define the time/frequency *centers* and *widths*

$$\mu_t \triangleq \frac{1}{E} \int_{-\infty}^{\infty} t \, |b(t)|^2 \, \mathrm{d}t, \qquad \sigma_t \triangleq \sqrt{\frac{1}{E} \int_{-\infty}^{\infty} (t - \mu_t)^2 \, |b(t)|^2 \, \mathrm{d}t}$$

$$\mu_\Omega \triangleq \frac{1}{E} \int_{-\infty}^{\infty} \Omega \, |B_{\mathsf{c}}(j\Omega)|^2 \frac{\mathrm{d}\Omega}{2\pi}, \quad \sigma_\Omega \triangleq \sqrt{\frac{1}{E} \int_{-\infty}^{\infty} (\Omega - \mu_\Omega)^2 \, |B_{\mathsf{c}}(j\Omega)|^2 \frac{\mathrm{d}\Omega}{2\pi}}$$



- For ST-CTFT basis element $b_{\Omega_0,\tau_0}(t) = \frac{1}{\sqrt{2\pi}} w(t - \tau_0) e^{j\Omega_0 t}$ with real, symmetric, unit-energy $w(t)$, one can show

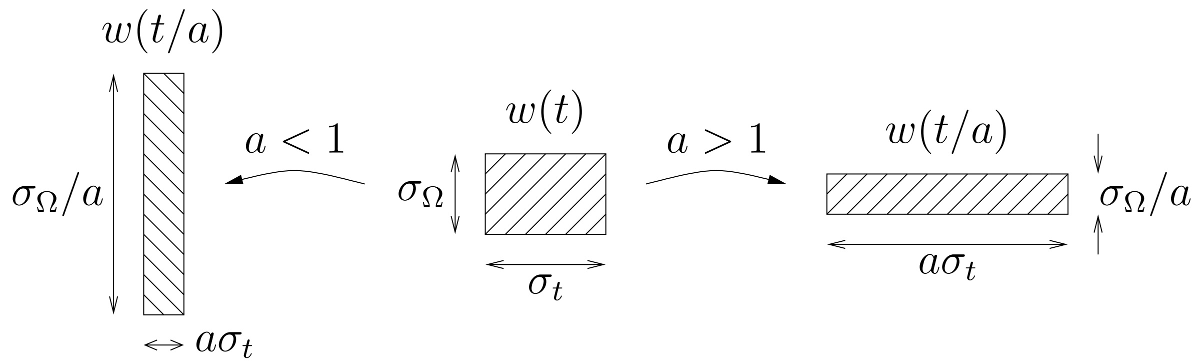
$$\mu_t = \tau_0, \qquad \sigma_t = \sqrt{\int_{-\infty}^{\infty} t^2 \, |w(t)|^2 \, \mathrm{d}t}$$

$$\mu_\Omega = \Omega_0, \quad \sigma_\Omega = \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} \Omega^2 \, |W_{\mathsf{c}}(j\Omega)|^2 \, \mathrm{d}\Omega}.$$

  So $b_{\Omega_0,\tau_0}(t)$ is centered at (τ_0, Ω_0) and has widths $(\sigma_t, \sigma_\Omega)$ that depend on the shape of the window $w(t)$.

Time/Frequency Uncertainty Principle:

- Stretching $w(t)$ in one domain compresses it in the other:



  implying a tradeoff between time & frequency resolution.

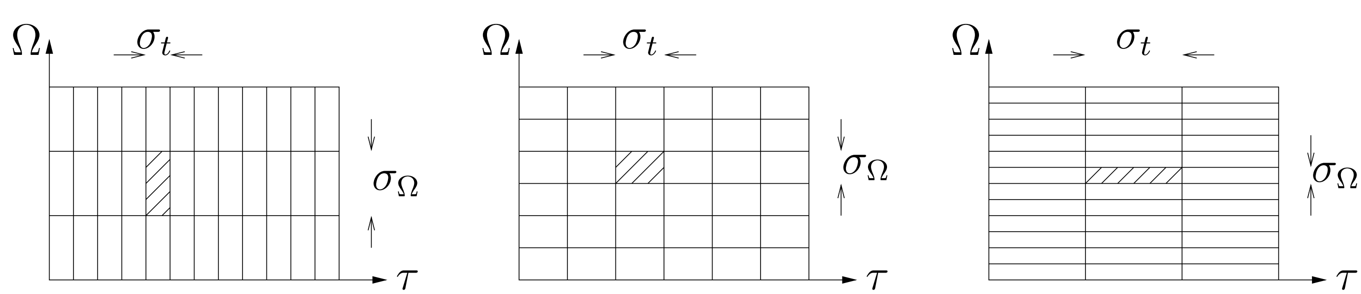
- The *time-frequency uncertainty principle* says, for any window $w(t)$, and thus any $b(t)$,

$$\sigma_t\sigma_\Omega \geq \tfrac{1}{2},$$

  where equality is reached by a Gaussian window, i.e.,

$$w(t) = (2c/\pi)^{\frac{1}{4}} e^{-ct^2} \text{ for any } c > 0.$$

- Together, the ST-CTFT basis functions $\{b_{\Omega,\tau}(t)\}$ form a "tiling" of the time-frequency plane that looks like



  depending on the window $w(t)$ used to construct $b_{\Omega,\tau}(t)$. The tile *area* can be no smaller than $1/2$.

Short-time DTFT:

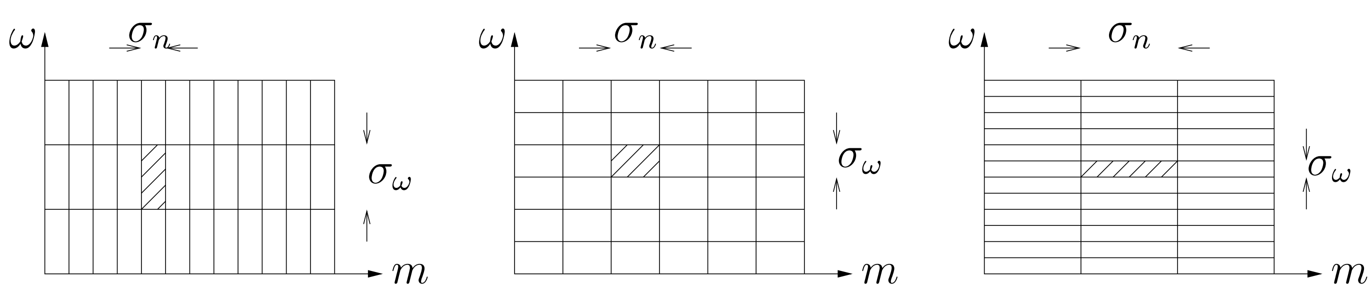
- The ST-CTFT applies to continuous-time signals $x(t)$.
- For discrete-time signals $x[n]$, we have the ST-DTFT:

$$X(e^{j\omega}, m) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_{m=-\infty}^{\infty} X(e^{j\omega}, m)w[n-m]e^{j\omega n}\,\mathrm{d}\omega,$$

where we have assumed $w[n] \in \mathbb{R}$ and $\sum_n w[n]^2 = 1$.

- The ST-DTFT constructs/deconstructs $x[n]$ using basis sequences $b_{\omega,m}[n] = \frac{1}{\sqrt{2\pi}}w[n-m]e^{j\omega n}$ for various ω, m.
- The ST-DTFT is very similar to the ST-CTFT, except that the time and shift domains are discrete and the frequency domain is 2π-periodic.



The definitions of σ_n and σ_ω in the figures are similar to those of σ_t and σ_Ω from earlier.

Short-time DFT:

- Most commonly, the ST-DTFT is used for *analysis*: we want to compute $|X(e^{j\omega}, m)|$ but we don't care about reconstructing $x[n]$ from $X(e^{j\omega}, m)$. (We'll talk about reconstruction later, in the context of filterbanks.)

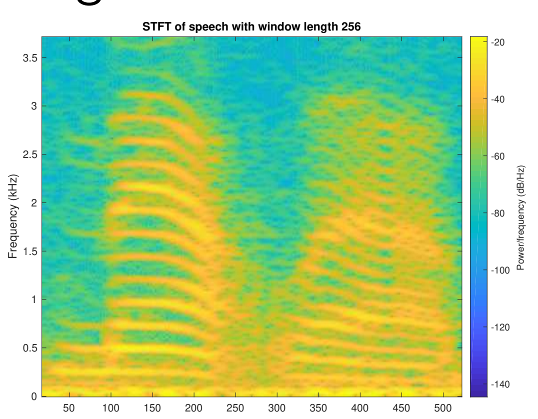
- For practicality, ω is sampled at N frequencies $\{\frac{2\pi k}{N}\}_{k=0}^{N-1}$ and m is downsampled by some factor $M \geq 1$:

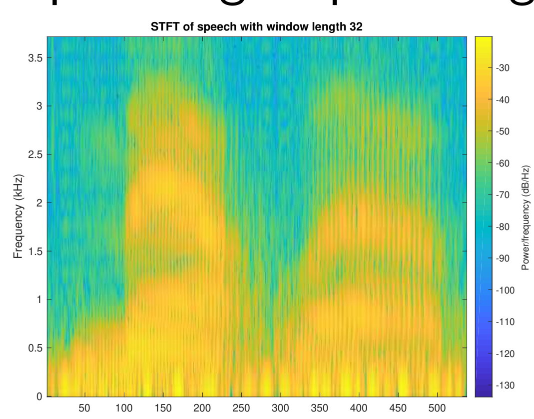
$$X[k,l] \triangleq X(e^{j\frac{2\pi}{N}k}, lM) = \sum_{n=-\infty}^{\infty} x[n]w[n-lM]e^{-j\frac{2\pi}{N}kn},$$

  giving the ST-DFT.

- As before, the length of the window $\{w[n]\}$ determines the time resolution and frequency resolution.

- Matlab's "`spectrogram`" computes & plots the ST-DFT magnitude. Here's an example using a speech signal:





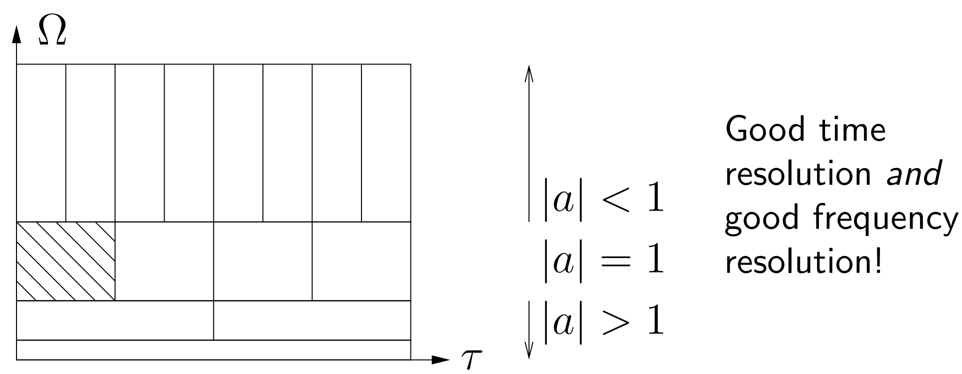
Left plot shows formant at $\approx$ 200 Hz.

Right plot shows 5 ms periodicity, which is consistent.

  - With a long window (left), we can see individual harmonics.
  - With a short window (right), we can see individual oscillations.

A Better Tiling?:

- The ST-CTFT uses *time & frequency shifts* of a single window function: $b_{\Omega,\tau}(t) = \frac{1}{\sqrt{2\pi}} w(t-\tau)e^{j\Omega t}$.
  - The choice of the window $w(t)$ involves a *tradeoff* between time & frequency resolution.
- What if we instead used *time shifts and stretches* of a single function $\psi(t)$? This would yield a tiling like



- To understand why this latter tiling may be better. . .
  - Say we wanted to distinguish between sinusoids at frequencies 1000 & 1001 Hz. After 1 sec, they differ by 1 period $\Rightarrow$ orthogonal $\Rightarrow$ easy to tell apart.
  - Now consider sinusoids at freqs 1 Hz & 1.001 Hz. After *1000 sec*, they differ by 1 period $\Rightarrow$ orthogonal $\Rightarrow$ easy to tell apart.
  - So, as the frequency Ω *decreases*, we require a *longer* observation period for good frequency resolution.
  - The above tiling does this!
  - It also provides good time resolution via the skinny top tiles!

The Continuous Wavelet Transform:

- The tiling on the previous page results from the *continuous wavelet transform* (CWT)

$$X_{\mathsf{cwt}}(a,\tau) = \int_{-\infty}^{\infty} x(t)\psi_{a,\tau}^*(t)\,\mathrm{d}t, \qquad \tau \in \mathbb{R}, a > 0$$

$$x(t) = \frac{1}{C_\psi}\int_{0_+}^{\infty}\int_{-\infty}^{\infty} X_{\mathsf{cwt}}(a,\tau)\psi_{a,\tau}(t)\frac{\mathrm{d}\tau\,\mathrm{d}a}{a^2}$$

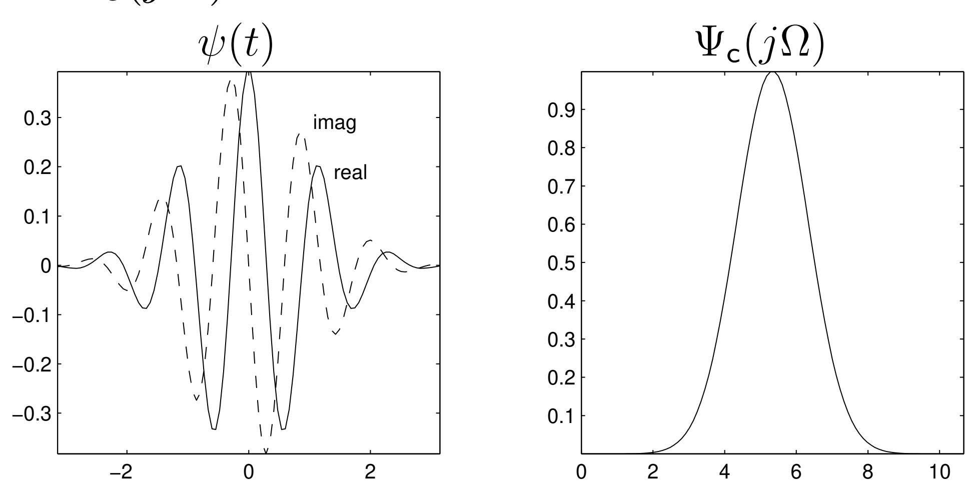
  where $\psi_{a,\tau}(t)$ is an a-stretched and τ-shifted copy of some "mother wavelet" $\psi(t)$:

$$\psi_{a,\tau}(t) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-\tau}{a}\right) \text{ with } \begin{cases} \int_{-\infty}^{\infty}\psi(t)\,\mathrm{d}t = 0 \\ C_\psi \triangleq \int_{-\infty}^{\infty}\frac{|\Psi_{\mathsf{c}}(j\Omega)|^2}{|\Omega|}\,\mathrm{d}\Omega < \infty \end{cases}$$

- A classic example is the Morlet wavelet, which uses a modulated Gaussian pulse:

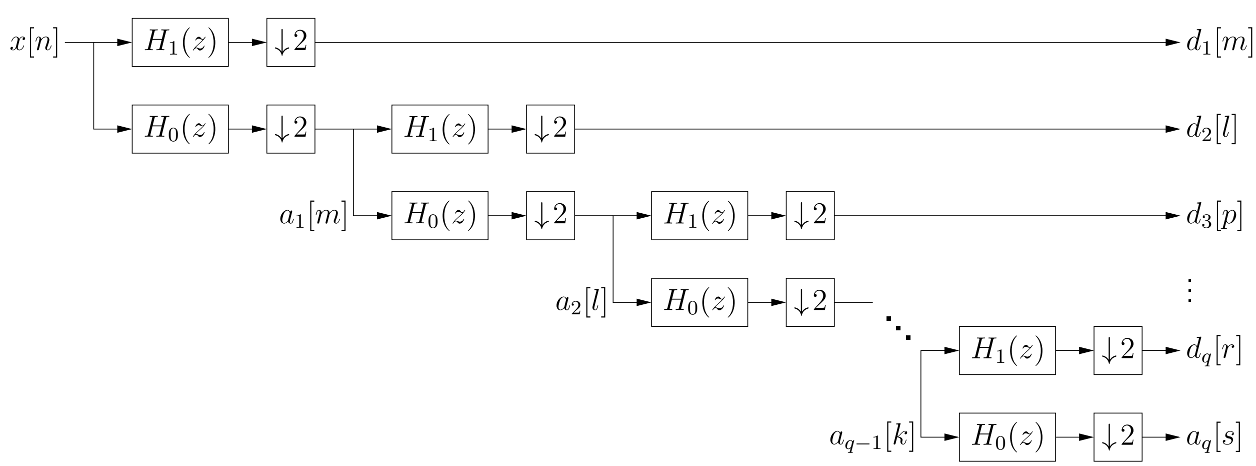
$$\psi(t) = \frac{1}{\sqrt{2\pi}}e^{j\Omega_0 t}e^{-t^2/2}, \quad \Omega_0 = \pi\sqrt{\frac{2}{\ln 2}}$$

$$\Psi_{\mathsf{c}}(j\Omega) = e^{-(\Omega-\Omega_0)^2/2}$$



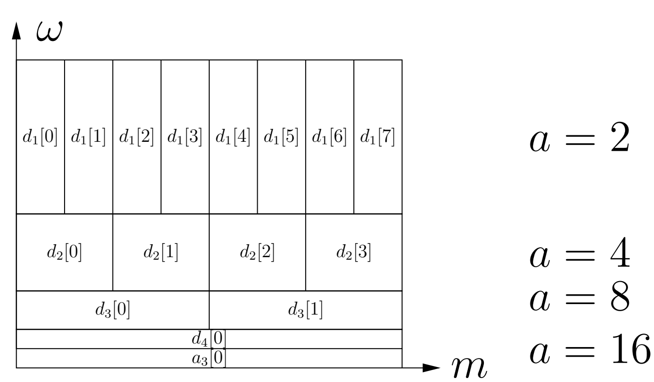
The Discrete Wavelet Transform (DWT):

- The CWT is not very practical since the time t, stretch a, and shift τ variables are all *continuous valued*.
- To be practical, we'd like a wavelet transform that operates on *discrete-time* signals $x[n]$ and uses *discrete-valued* stretches and shifts, i.e., a DWT.
- We can do this by restricting to "dyadic" stretches $a \in \{2^q : q \in \mathbb{Z}_+\}$ and using an iterated filterbank



with appropriate highpass $H_1(z)$ and lowpass $H_0(z)$.

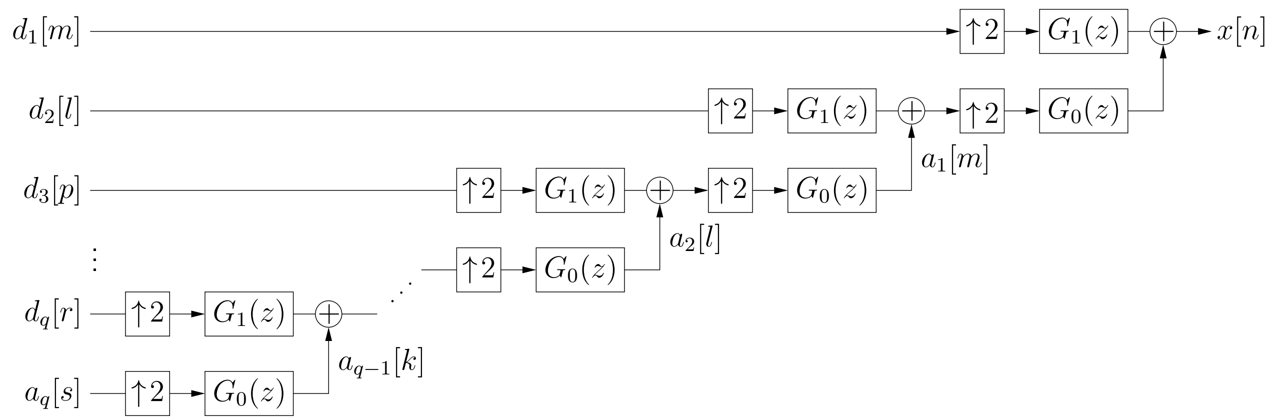
- This yields the tiling (for $Q = 4$ levels and length $N = 16$):



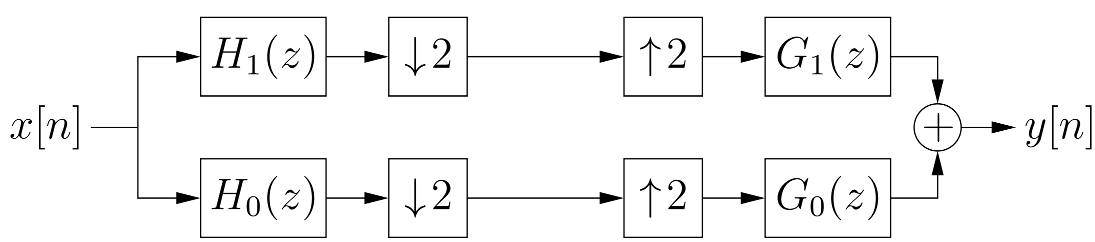
- Note that the # DWT outputs = # DWT inputs!

The Discrete Wavelet Transform (cont.):

- For the inverse DWT (i.e., reconstruction of $x[n]$), we use an iterated synthesis filterbank



- Notice that the inverse DWT will perfectly reconstruct $x[n]$ if and only if the two-channel filterbank



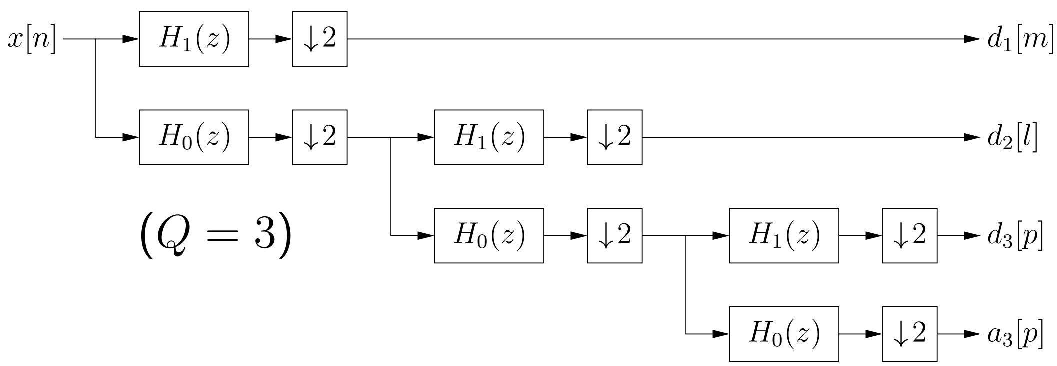
  is itself perfect reconstructing, i.e., if $y[n] = x[n]$.

- We will design perfect-reconstruction (PR) filters $\{H_0(z), H_1(z), G_0(z), G_1(z)\}$ in the next note packet.

- With certain PR filters, we get an *orthogonal* DWT, i.e., $\boxed{\boldsymbol{c} = \boldsymbol{W}\boldsymbol{x}}$, where the DWT matrix $\boldsymbol{W}$ is real-valued and unitary. The inverse DWT then becomes $\boxed{\boldsymbol{x} = \boldsymbol{W}^{\mathsf{T}}\boldsymbol{c}}$.

1D Orthogonal DWT Details:

- The orthogonal DWT transforms an N-length sequence $\{x[n]\}_{n=0}^{N-1}$ to N DWT coefficients using Q levels.
  - you can choose any $Q \in \{1, 2, \ldots, \log_2 N\}$
  - you can choose the filter designs, e.g., orthogonal Daubechies `db1,db2,db3,...` (see next note packet)



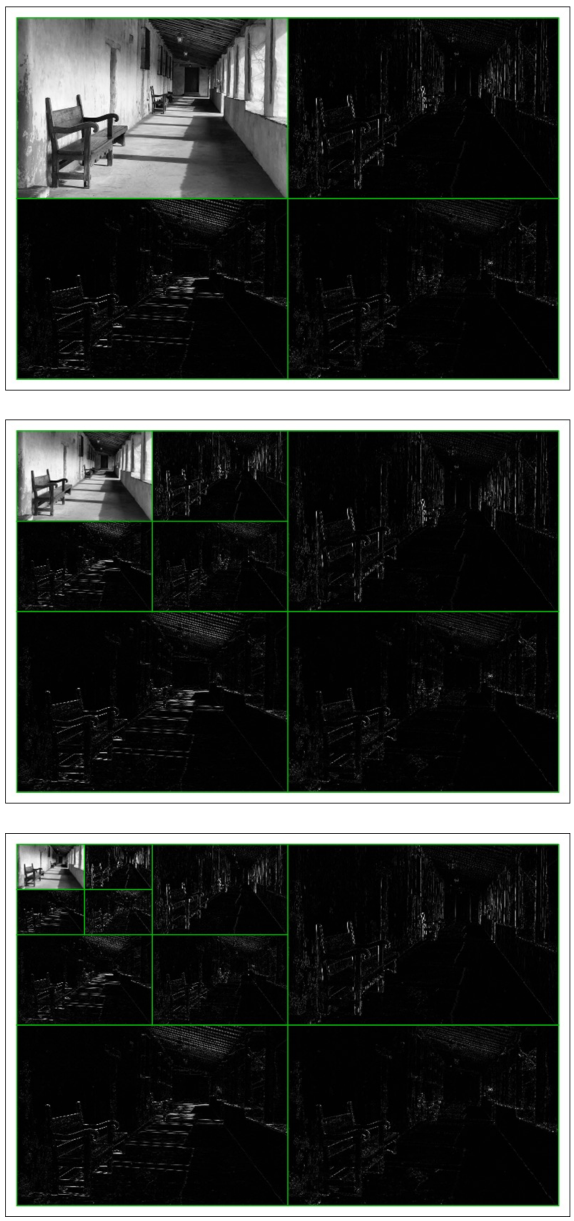
- The DWT coefficients are organized as follows:
  - At level $q = 1, ..., Q$, there are $\frac{N}{2^q}$ "detail" coefficients $\{d_q[\cdot]\}$.
  - There are $\frac{N}{2^Q}$ "approximation" coefficients $\{a_Q[p]\}$.
- When writing "$\boldsymbol{c} = \boldsymbol{W}\boldsymbol{x}$," it is typical to use $\boldsymbol{c} = \begin{bmatrix} \boldsymbol{a}_Q \\ \boldsymbol{d}_Q \\ \boldsymbol{d}_{Q-1} \\ \vdots \\ \boldsymbol{d}_1 \end{bmatrix}$, where subvector $\boldsymbol{a}_Q$ is constructed from $\{a_Q[p]\}$ and each subvector $\boldsymbol{d}_q$ is constructed from $\{d_q[\cdot]\}$.
- Matlab example:

```
dwtmode('per');                   % set boundary conditions
[c,len] = wavedec(x,Q,'db2'); % DWT
x = waverec(c,len,'db2');     % inverse DWT
```

where "`len`" lists the lengths of the subvectors in c.

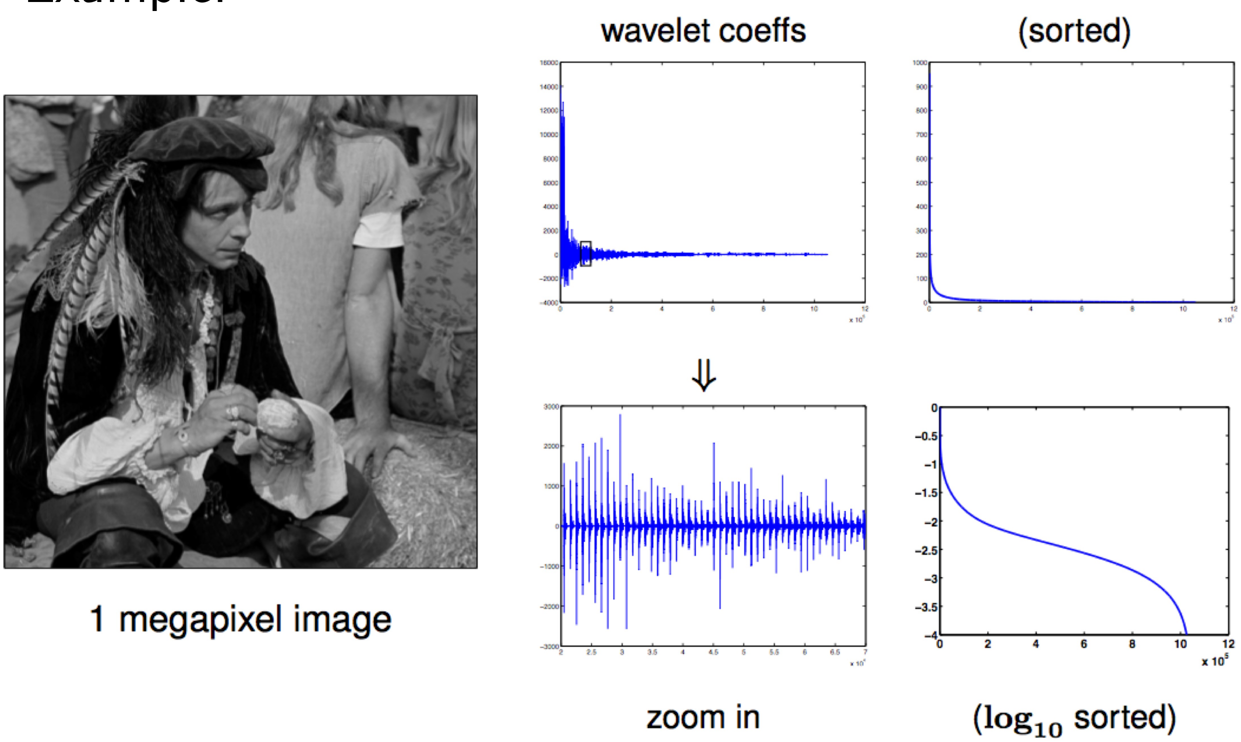
2D Orthogonal DWT::

- An image is a 2D signal, often represented as a matrix $\boldsymbol{X}$.
- The 2D-DWT takes a 1D-DWT of each column and a 1D-DWT of each row of the result: $\boldsymbol{C} = (\boldsymbol{W}\boldsymbol{X})\boldsymbol{W}^{\mathsf{T}}$.
  - The transpose is explained by $\boldsymbol{C}^{\mathsf{T}} = \boldsymbol{W}(\boldsymbol{W}\boldsymbol{X})^{\mathsf{T}}$.
- The 2D-DWT at levels $Q = 1, 2, 3$ is shown below:



Wavelet-domain Sparsity:

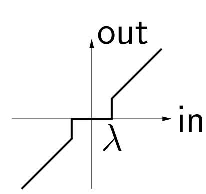
- Importantly, the DWT coefficients of natural signals are *sparse*, i.e., mostly zero-valued.
  - "natural" means, e.g., images, speech, audio, video.
- Example:



In this example, 99% of the signal energy is captured by only 2.5% of the DWT coefficients!

- One application is *(lossy) compression*. To do this, we *threshold* the DWT coefficients, keeping only the largest in magnitude. This is used in JPEG, MPEG, etc.

$$\widehat{\boldsymbol{x}} = \boldsymbol{W}^{\mathsf{T}}\text{thresh}(\boldsymbol{W}\boldsymbol{x}, \lambda), \qquad \text{thresh}(\cdot, \lambda) =$$



Sparsity-based Denoising:

- Suppose that we measure a corrupted version $\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{n}$ of a signal $\boldsymbol{x}$ and we'd like to remove the noise $\boldsymbol{n}$.
- Taking the DWT of the measurements yields

$$\boldsymbol{W}\boldsymbol{y} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{W}\boldsymbol{n} = \boldsymbol{c} + \boldsymbol{W}\boldsymbol{n}.$$

- Due to the orthogonality of $\boldsymbol{W}$ and randomness of $\boldsymbol{n}$, $\boldsymbol{W}\boldsymbol{n}$ has (on average) equal energy in each coefficient. But $\boldsymbol{c}$ has most of its energy in only a *few* coefficients.
- To approximately recover $\boldsymbol{x}$, we can use wavelet-domain thresholding with $\lambda \approx 3.5 \times$ the noise standard-deviation:

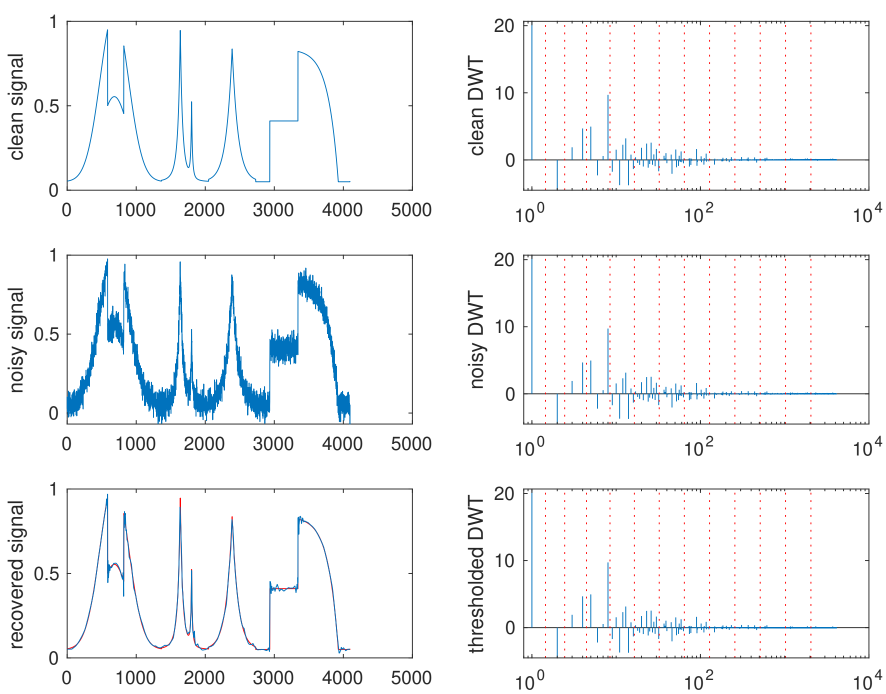
$$\begin{aligned}\widehat{\boldsymbol{x}} &= \boldsymbol{W}^{\mathsf{T}}\text{thresh}(\boldsymbol{W}\boldsymbol{y}, \lambda) \\ &= \boldsymbol{W}^{\mathsf{T}}\text{thresh}(\boldsymbol{c} + \boldsymbol{W}\boldsymbol{n}, \lambda) \approx \boldsymbol{W}^{\mathsf{T}}\boldsymbol{c} = \boldsymbol{x}\end{aligned}$$

- Example:

Note that the DTW coefs (right column) are plotting using a logarithmic horizontal axis.

The red dashed lines partition the vector $\boldsymbol{c}$ into its subvectors $\boldsymbol{a}_Q, \boldsymbol{d}_Q, \ldots, \boldsymbol{d}_1$.

Here $Q = \log_2 N = 12$.



Sparsity-based sub-Nyquist Sampling:

- Wavelet-domain sparsity can also be exploited for *sub-Nyquist sampling*.
- Suppose that we wish to recover an N-length signal $\boldsymbol{x}$ from M linear measurements $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$, where $M \ll N$.
  - Inpainting: $\boldsymbol{A}$ is random rows of identity matrix.
  - Medical/radar imaging: $\boldsymbol{A}$ is random rows of 2D-DFT matrix.
  - Compressive sensing: $\boldsymbol{A}$ has independent random entries.
- Say $\boldsymbol{x}$ has K-sparse DWT coefs $\boldsymbol{c}$ (so that $\boldsymbol{x} = \boldsymbol{W}^\mathsf{T}\boldsymbol{c}$).
- To approximately recover $\boldsymbol{x}$ from $\boldsymbol{y}$, use $\widehat{\boldsymbol{x}} = \boldsymbol{W}^\mathsf{T}\widehat{\boldsymbol{c}}$ with

$$\widehat{\boldsymbol{c}} = \arg\min_{\boldsymbol{c}} \|\boldsymbol{c}\|_0 \text{ s.t. } \boldsymbol{y} = \boldsymbol{A}\boldsymbol{W}^\mathsf{T}\boldsymbol{c},$$

or its convex relaxation

$$\widehat{\boldsymbol{c}} = \arg\min_{\boldsymbol{c}} \|\boldsymbol{c}\|_1 \text{ s.t. } \boldsymbol{y} = \boldsymbol{A}\boldsymbol{W}^\mathsf{T}\boldsymbol{c}.$$

Good recovery with only $M \gtrsim K \log N$ measurements!

- Inpainting example: (taken from here)

original

30% sampled

recovered